

AI Engineering (Week 1)

aie26

Topic 04 : Exploratory Data Analysis

Part 01 : EDA Pass1

Preparation

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Data Handling

Exploring Data 1

Exploring Data 2

Building Models

Autumn Semester, 2026

Outline

- EDA Process
- Datasets = Tips, Titanic and Algae Blooms
- EDA Pass 1

Wrap up

AI Engineering (Week 4)

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Motivating Example

Preparation

Data Handling

Exploring Data 1

Exploring Data 2

Building Models

Prediction

Regression
1

Classification
1

Regression
2

Classification
2

Clustering

Wrap up

Outline

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Acknowledgment

A big thanks to Dr Kieran Murphy, who provided many of the slides for today's lecture.

Introduction

Exploratory Data Analysis (EDA)

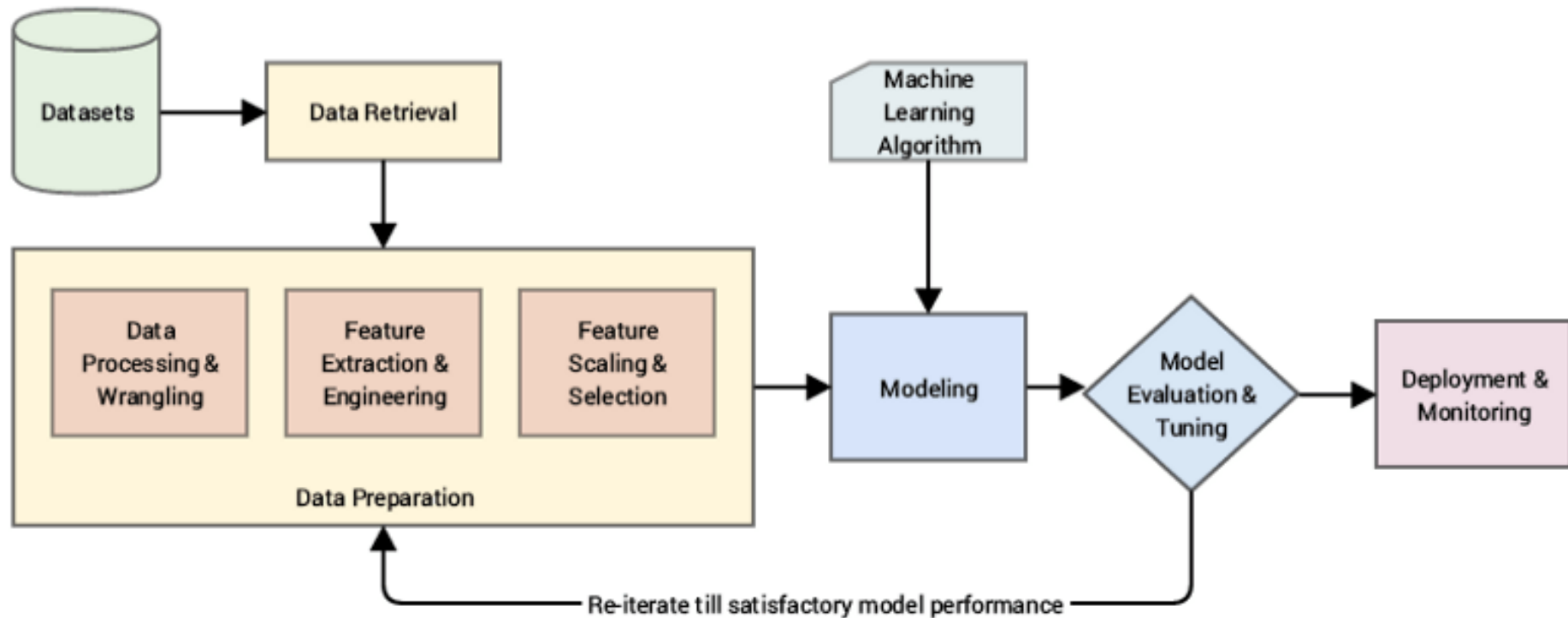
Aim

To understand and summarise a dataset to ensure that the features which are feed to machine learning algorithms are refined and that the results are valid and can correctly interpreted.

Benefits

- Develop insight about the dataset and understanding of the underlying structure.
- Extract important parameters and relationships that hold between them.
- Test underlying assumptions.
- Identify issues that affect model performance — outliers, missing values.

Data Pipeline



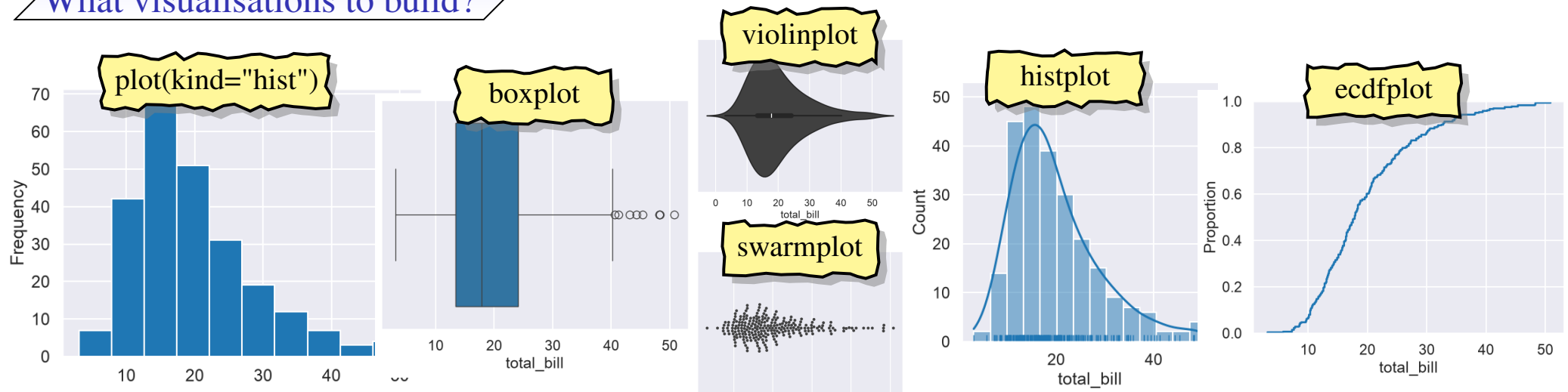
- Data preparation is the core of the machine learning pipeline (typical estimates >50% of the time/effort).
- EDA is the data processing and wrangling before models are built.
- EDA informs the feature extraction, engineering, transformation and selection.

The Bad News — ‘The curse of choice’

What questions to ask?

Dataset global questions: How many features? How many observations? What is the data type of each feature? Any null values? ... Feature specific questions: What is the distribution of each variable? Do there appear to be outliers? What features are related? ... Missing value questions: Are null value a result of the way data was recorded? Can we drop the rows with null values without it significantly affecting your analysis? Can we justify filling in the missing values with the mean or median for that variable? If the data is time-series data, can we fill the missing values with interpolation? Are there so many missing values for a variable that we should drop that variable from the dataset? ... Outlier questions: Why are outliers present? Do the outliers represent real observations (i.e. not errors)? Should we exclude these observations? If not, should we winsorise the values? ... Correlations/Relationships questions: Which variables are most correlated with your target variable? (If applicable) Is there multicollinearity? (Two features that have a correlation > 0.8) How will this affect your model? Do you have variables that represent the same information? Can one be dropped? ...

What visualisations to build?



Have a plan, be selective, understand strengths/weaknesses of metrics/visualisations

Terminology / Notation

$n + 1$ columns / variables

X

n features / attributes / dimensions

y
target

m observations /
instances /
cases / rows

PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Survived
1	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	0
2	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	1
3	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	1
4	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S	1
5	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S	0
6	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q	0
7	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S	0
8	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S	0
9	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S	1
10	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C	1
11	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000	G6	S	1

$\mathbf{x}_j$

$\mathbf{x}^{(i)}$

- A labeled dataset consists of m rows $\times$ $(n + 1)$ columns / variables.
- Use bold to represent vectors and matrices.
- Use subscripts to indicate particular **feature / attribute / column** $\mathbf{x}_j$
- Use superscript in parenthesis to indicate particular **observation / instance/ case / row** $\mathbf{x}^{(i)}$
- So $x_j^{(i)}$ (or $x_{i,j}$) is the i -th observation in the j -th feature $x_j^{(i)}$

Example Datasets

We will use a few datasets today to illustrate the various features:

Tips

- Small dataset of total bills, and tips for different servers with gender, day, time and group size.
- Clean, no missing values, some outliers.
- Task: exploratory data analysis

Titanic

- Classic dataset with passenger information for the Titanic's fatal voyage, and whether they survived.
- Has missing values and information rich text fields (Name, ticket number).
- Task: classification — predict whether a passenger survived.

Algae Blooms

- Water quality study where samples were taken from different rivers over time.
- Recorded levels of (seven) chemical substances and population of (six) algae species and other information on the sample conditions.
- Task: regression — predict algae population level (7 separate populations).

Tips **dataset**

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
5	25.29	4.71	Male	No	Sun	Dinner	4
6	8.77	2.00	Male	No	Sun	Dinner	2
7	26.88	3.12	Male	No	Sun	Dinner	4
8	15.04	1.96	Male	No	Sun	Dinner	2
9	14.78	3.23	Male	No	Sun	Dinner	2

No target column, so mainly just an exploratory data analysis problem. But questions of interest:

- How do factors **sex**, **smoker**, **day**, **time**, or **size** affect tip / percentage tip?
- Does **size** vary with **day**, **time**, **smoker**?

But some questions don't make sense

- What is the relationship between **sex** and **smoker**? — why should they be related?

This is the downside of automatic EDA tools such as **pandas-profiling** — you will drown in statistics / charts.

Algae Blooms dataset

	Season	Size	Speed	max_pH	min_O2	mean_Cl	mean_NO3	mean_NH4	mean_oPO4	mean_PO4	mean_Chlor	a1	a2	a3	a4	a5
0	winter	small	medium	8.00	9.8	60.800	6.238	578.00000	105.00000	170.00000	50.000	0.0	0.0	0.0	0.0	34.2
1	spring	small	medium	8.35	8.0	57.750	1.288	370.00000	428.75000	558.75000	1.300	1.4	7.6	4.8	1.9	6.7
2	autumn	small	medium	8.10	11.4	40.020	5.330	346.66699	125.66700	187.05701	15.600	3.3	53.6	1.9	0.0	0.0
3	spring	small	medium	8.07	4.8	77.364	2.302	98.18200	61.18200	138.70000	1.400	3.1	41.0	18.9	0.0	1.4
4	autumn	small	medium	8.06	9.0	55.350	10.416	233.70000	58.22200	97.58000	10.500	9.2	2.9	7.5	0.0	7.5
5	winter	small	high	8.25	13.1	65.750	9.248	430.00000	18.25000	56.66700	28.400	15.1	14.6	1.4	0.0	22.5
6	summer	small	high	8.15	10.3	73.250	1.535	110.00000	61.25000	111.75000	3.200	2.4	1.2	3.2	3.9	5.8
7	autumn	small	high	8.05	10.6	59.067	4.990	205.66701	44.66700	77.43400	6.900	18.2	1.6	0.0	0.0	5.5
8	winter	small	medium	8.70	3.4	21.950	0.886	102.75000	36.30000	71.00000	5.544	25.4	5.4	2.5	0.0	0.0
9	winter	How well can we predict the (7) different algae population levels using water sample information?														2.9
10	spring	small	high	7.70	10.2	8.000	1.527	21.57100	12.75000	20.75000	0.800	16.6	0.0	0.0	0.0	1.2
11	summer	small	high	7.45	11.7	8.690	1.588	18.42900	10.66700	19.00000	0.600	32.1	0.0	0.0	0.0	0.0
12	winter	small	high	7.74	9.6	5.000	1.223	27.28600	12.00000	17.00000	41.000	43.5	0.0	2.1	0.0	1.2
13	summer	small	high	7.72	11.8	6.300	1.470	8.00000	16.00000	15.00000	0.500	31.1	1.0	3.4	0.0	1.9
14	winter	small	high	7.90	9.6	3.000	1.448	46.20000	13.00000	61.60000	0.300	52.2	5.0	7.8	0.0	4.0
15	autumn	small	high	7.55	11.5	4.700	1.320	14.75000	4.25000	98.25000	1.100	69.9	0.0	1.7	0.0	0.0
16	winter	small	high	7.78	12.0	7.000	1.420	34.33300	18.66700	50.00000	1.100	46.2	0.0	0.0	1.2	0.0
17	spring	small	high	7.61	9.8	7.000	1.443	31.33300	20.00000	57.83300	0.400	31.8	0.0	3.1	4.8	7.7
18	summer	small	high	7.35	10.4	7.000	1.718	49.00000	41.50000	61.50000	0.800	50.6	0.0	9.9	4.3	3.6
19	spring	small	medium	7.79	3.2	64.000	2.822	8777.59961	564.59998	771.59998	4.500	0.0	0.0	0.0	44.6	0.0

Titanic dataset

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0 1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1 2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2 3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3 4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4 5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5 6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6 7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7 8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S
8 9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9 10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C
10 11	1	2	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	DR 0540	16.7000	G6	S
11 12										103	S
12 13	0	3	Saundercock, Mr. William Henry	male	20.0	0	0	A/5. 2151	8.0500	NaN	S
13 14	0	3	Andersson, Mr. Anders Johan	male	39.0	1	5	347082	31.2750	NaN	S
14 15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14.0	0	0	350406	7.8542	NaN	S
15 16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55.0	0	0	248706	16.0000	NaN	S
16 17	0	3	Rice, Master. Eugene	male	2.0	4	1	382652	29.1250	NaN	Q

How well can we predict a passenger's survival using information at time of departure?

Before we start ... Loading libraries

We start by loading in the core data science modules...

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

matplotlib is an excellent visualisation library but some plots need additional configuration. seaborn sits above matplotlib and has a collection of visualisations optimised for statistical analysis. ...

```
import seaborn as sns
```

Next, we import some statistical modules ...

```
import scipy.stats as stats
import statsmodels.api as sm
import pingouin as pg
```

`scipy.stats` has a large number of distributions, parametric and nonparametric statistical tests, and descriptive statistics.
`statsmodels` is more focused on estimating statistical models.
`pingouin` overlaps with bits of `scipy.stats` and `statsmodels` but generates more details and nicer visualisations.

Finally we set options ...

```
plt.style.use("seaborn-v0_8-darkgrid")
```

Before we start ... auto EDA using ydata-profiling

```
from pandas_profiling import ProfileReport
profile = ProfileReport(df, title="Tips Report", html={"style": {"full_width": True}}, sort="None")
profile
```

Summarize dataset: 100%  21/21 [00:05<00:00, 4.00it/s, Completed]

Generate report structure: 100%  1/1 [00:02<00:00, 2.78s/it]

Render HTML: 100%  1/1 [00:00<00:00, 1.72it/s]

ydata-profiling is nice, but see how slow it is on this tiny dataset. What would happen if we had 100K rows x 100 columns?

Tips Report

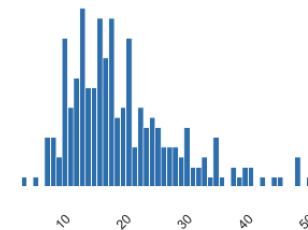
Overview Variables Interactions Correlations Missing values Sample Duplicate rows

Variables

total_bill

Real number ($\mathbb{R}_{\geq 0}$)

Distinct	229	Mean	19.78594262
Distinct (%)	93.9%	Minimum	3.07
Missing	0	Maximum	50.81
Missing (%)	0.0%	Zeros	0
Infinite	0	Zeros (%)	0.0%
Infinite (%)	0.0%	Memory size	1.9 KiB

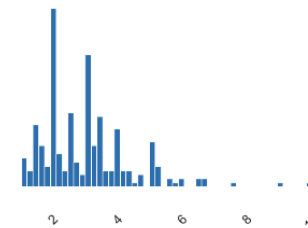


Toggle details

tip

Real number ($\mathbb{R}_{\geq 0}$)

Distinct	123	Mean	2.998278689
Distinct (%)	50.4%	Minimum	1
Missing	0	Maximum	10
Missing (%)	0.0%	Zeros	0
Infinite	0	Zeros (%)	0.0%
Infinite (%)	0.0%	Memory size	1.9 KiB



Before we start ... zero-code EDA using dtale

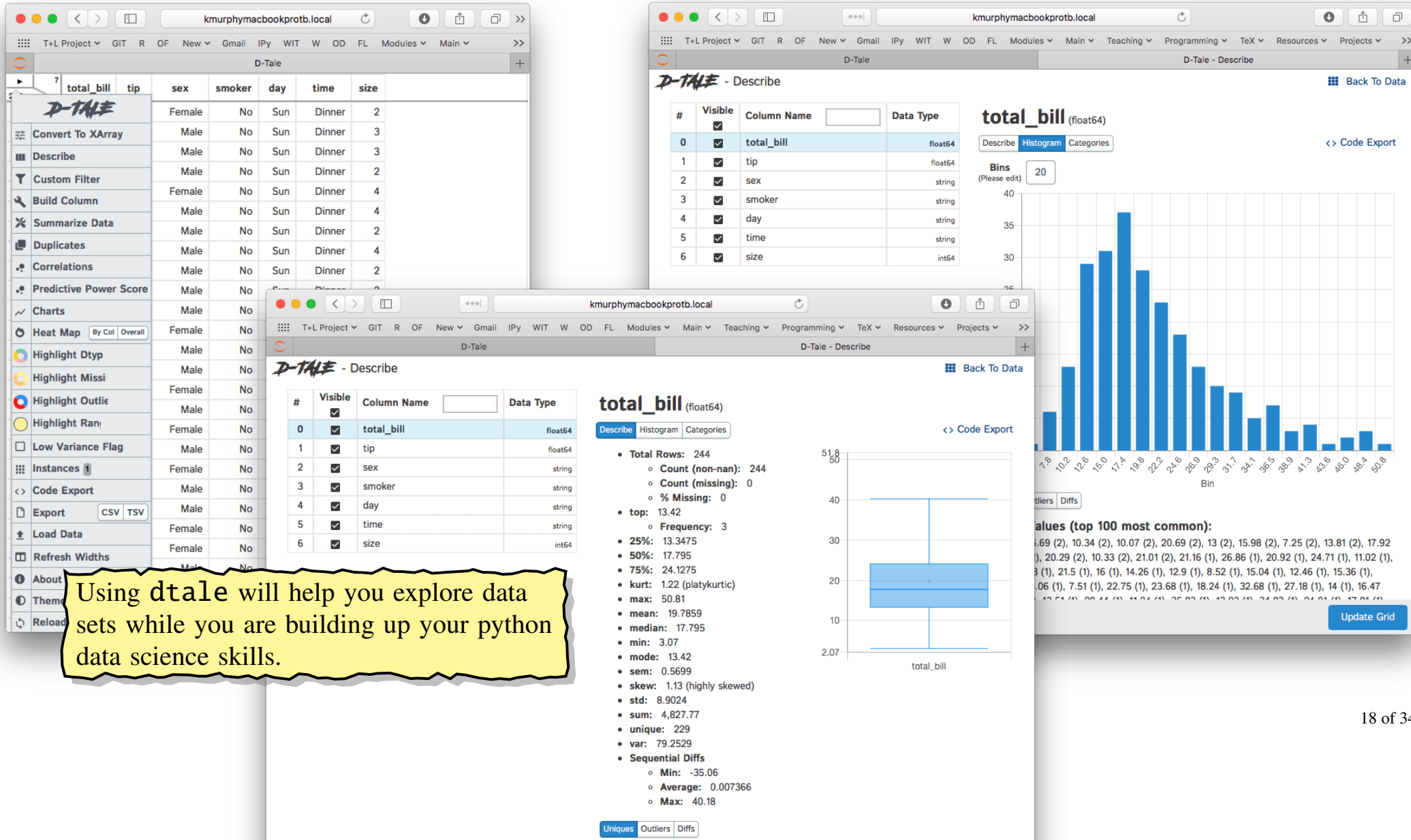
Well, almost zero code....

```
import pandas as pd
import dtale

# Read the Tips data into a dataframe, check it looks OK
df = pd.read_csv('tips.csv')
df.head()

# Run dtale to visualize the structure of the dataframe
dtale.show(df)
```

Before we start ... zero-code EDA using dtale



First Pass — Load Dataset and Initial Clean

- Load dataset
- Check variables names
- Verify variable types
- Identify (and possibly address) missing values

Tips — Load

```
df = pd.read_csv("data/tips.csv")
print(df.shape)
df.head(10)
```

(244, 7)

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
5	25.29	4.71	Male	No	Sun	Dinner	4
6	8.77	2.00	Male	No	Sun	Dinner	2
7	26.88	3.12	Male	No	Sun	Dinner	4
8	15.04	1.96	Male	No	Sun	Dinner	2
9	14.78	3.23	Male	No	Sun	Dinner	2

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex        244 non-null    str
3   smoker     244 non-null    str
4   day        244 non-null    str
5   time       244 non-null    str
6   size       244 non-null    int64
dtypes: float64(2), int64(1), str(4)
memory usage: 13.5 KB
```

Issue: categorical data treated as object (string).

Tips — Fix Data Types

```
df.sex.unique()
```

```
<StringArray>  
['Female', 'Male']  
Length: 2, dtype: str
```

```
df.sex = pd.Categorical(df.sex)  
df.sex.unique()
```

```
['Female', 'Male']  
Categories (2, str): ['Female', 'Male']
```

```
df.day.unique()
```

```
<StringArray>  
['Sun', 'Sat', 'Thur', 'Fri']  
Length: 4, dtype: str
```

```
df.day = pd.Categorical(df.day, categories=['Thur', 'Fri', 'Sun', 'Sat'], ordered=True)  
df.day.unique()
```

```
['Sun', 'Sat', 'Thur', 'Fri']  
Categories (4, str): ['Thur' < 'Fri' < 'Sun' < 'Sat']
```

```
df.smoker.unique()
```

```
<StringArray>  
['No', 'Yes']  
Length: 2, dtype: str
```

```
df.smoker = pd.Categorical(df.smoker)  
df.smoker.unique()
```

```
['No', 'Yes']  
Categories (2, str): ['No', 'Yes']
```

Tips — fix datatypes

```
df.time = pd.Categorical(df.time, categories=['Lunch', 'Dinner'], ordered=True)
df.time.unique()
```

```
['Dinner', 'Lunch']
Categories (2, str): ['Lunch' < 'Dinner']
```

```
df.info()
```

Converting to category will:

- Simplify visualisation (order can be preserved).
 - Reduce memory usage (not that big a deal for us).
 - Speed up I/O (depending on file format).
- ⇒ Convert to category is a bigger deal for features where the levels have an order.

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 244 entries, 0 to 243
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	total_bill	244 non-null	float64
1	tip	244 non-null	float64
2	sex	244 non-null	category
3	smoker	244 non-null	category
4	day	244 non-null	category
5	time	244 non-null	category
6	size	244 non-null	int64

```
dtypes: category(4), float64(2), int64(1)
```

```
memory usage: 7.2 KB
```

Titanic — load

- Dataset is split into two parts:

- `train.csv` — 891 rows with `Survived` column, used in EDA and model training.
- `test.csv` — 418 rows without the `Survived` column, used in competition scoring.

```
df = pd.read_csv("data/train.csv")
print(df.shape)
df.head(25)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	str
4	Sex	891 non-null	str
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	str
9	Fare	891 non-null	float64
10	Cabin	204 non-null	str
11	Embarked	889 non-null	str

```
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

- We could convert `Sex` or `Embarked`, to a category, but since their levels are not ordered there is no big advantage.
- We don't want to convert `Name`, `Ticket` and `Cabin` since we want to perform further text processing on these columns. For example, extracting title (Capt, Mr, Miss, etc.) out of `Name`.
- We have missing values (**that are plausibly linked to target**) that we need to deal with.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
											C123	S
											NaN	S
											NaN	Q
											E46	S
											NaN	S
											NaN	S
											NaN	C
											G6	S
											C103	S
											NaN	S
											NaN	S
											NaN	S
											NaN	Q
											NaN	S
											NaN	S
19	20	1	3	Massei, Mrs. Fatima	female	NaN	0	0	2649	7.2250	NaN	C
20	21	0	2	Fynney, Mr. Joseph J	male	35.0	0	0	239865	26.0000	NaN	S
21	22	1	2	Beesley, Mr. Lawrence	male	34.0	0	0	248698	13.0000	D56	S
22	23	1	3	McGowan, Miss. Anna "Annie"	female	15.0	0	0	330923	8.0292	NaN	Q
23	24	1	1	Sloper, Mr. William Thompson	male	28.0	0	0	113788	35.5000	A6	S
24	25	0	3	Palsson, Miss. Torborg Danira	female	8.0	3	1	349909	21.0750	NaN	S

Algae_Blooms – load

I

<https://archive.ics.uci.edu/ml/datasets/Coil+1999+Competition+Data>

Read instructions.txt, and download the *.data files.

UCI Machine Learning Repository
Center for Machine Learning and Intelligent Systems

Coil 1999 Competition Data Data Set
Download: [Data Folder](#), [Data Set Description](#)

Abstract: This data set is from the 1999 Computational Intelligence and Learning (COIL) competition. The data contains measurements of river chemical concentrations and algae densities.

Data Set Characteristics:	Multivariate	Number of Instances:	340	Area:	Physical
Attribute Characteristics:	Categorical, Real	Number of Attributes:	17	Date Donated	1999-09-09
Associated Tasks:	N/A	Missing Values?	No	Number of Web Hits:	52986

Source:

Original Owner:

ERUDIT
European Network for Fuzzy Logic and Uncertainty Modelling
<http://www.erudit.de/>

Donor:

Jens Strackeljan
Technical University Clausthal
Institute of Applied Mechanics
Graupenstr. 3, 38678 Clausthal-Zellerfeld, Germany
tmjs.*@*itn.tu-clausthal.de

Data Set Information:

This data comes from a water quality study where samples were taken from a river. The data includes chemical substances including: nitrogen in the form of nitrates, nitrites, and ammonia.

The competition involved the prediction of algal frequency distribution when the sample was taken, the river size and its flow velocity.

Index of /ml/machine-learning-databases/coil-mld

- [Parent Directory](#)
- [analysis.data](#)
- [coil.data.html](#)
- [coil.html](#)
- [eval.data](#)
- [instructions.txt](#)
- [r2](#)
- [results.data](#)
- [results.htm](#)
- [results.txt](#)

Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips SVN/1.7.14 Phusion_Passenger/4.0.53 mod_perl/2.0.11 Perl/v5.16.3 Server at archive.ics.uci.edu Port 443

Algae_Blooms — load (1st attempt)

Pandas function `pd.read_table`, is a more general function than `read_csv`.

```
df = pd.read_table('src/Analysis.txt')
print(df.shape)
df.head()
```

(199, 1)

	winter	small	medium	8.00000	9.80000	60.80000	6.23800	578.00000	105.00000	170.00000	50.00000	0.00000
								0.00000	0.00000	0.00000	34.20000	8.30000
0	spring	small	medium	8.35000	...							
1	autumn	small	medium	8.10000	1...							
2	spring	small	medium	8.07000	...							
3	autumn	small	medium	8.06000	...							
4	winter	small	high	8.25000	13...							

Two problems, first row was treated as column headers, and we need to specify the character(s) used to separate columns

Algae_Blooms — load (2nd attempt)

```
df = pd.read_table('src/Analysis.txt', sep=r'\s+', header=None)
print(df.shape)
df.head()
```

(200, 18)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
0	winter	small	medium	8.00000	9.80000	60.80000	6.23800	578.00000	105.00000	170.00000	50.00000	0.0	0.0	0.0	0.0	34.2	8.3	0.0
1	spring	small	medium	8.35000	8.00000	57.75000	1.28800	370.00000	428.75000	558.75000	1.30000	1.4	7.6	4.8	1.9	6.7	0.0	2.1
2	autumn	small	medium	8.10000	11.40000	40.02000	5.33000	346.66699	125.66700	187.05701	15.60000	3.3	53.6	1.9	0.0	0.0	0.0	9.7
3	spring	small	medium	8.07000	4.80000	77.36400	2.30200	98.18200	61.18200	138.70000	1.40000	3.1	41.0	18.9	0.0	1.4	0.0	1.4
4	autumn	small	medium	8.06000	9.00000	55.35000	10.41600	233.70000	58.22200	97.58000	10.50000	9.2	2.9	7.5	0.0	7.5	4.1	1.0

- Now, notice that the number of data rows changed from 199 to 200 since the first row is now counted as a data row. And now we are using default columns names.
- The "\s+" matches one or more spaces. This is an example of a regex.
- We need to name the columns.

Algae_Blooms — load (3rd attempt)

IV

```
names = ('Season', 'Size', 'Speed', 'max_pH', 'min_O2', 'mean_Cl', 'mean_NO3', 'mean_NH4', 'mean_oPO4',
         'mean_PO4', 'mean_Chlor', 'a1', 'a2', 'a3', 'a4', 'a5', 'a6', 'a7')
```

```
df = pd.read_table('src/Analysis.txt', sep=r'\s+', names=names)
```

```
print(df.shape)
```

```
df.head()
```

(200, 18)

	Season	Size	Speed	max_pH	min_O2	mean_Cl	mean_NO3	mean_NH4	mean_oPO4	mean_PO4	mean_Chlor	a1	a2	a3	a4	a5	a6
0	winter	small	medium	8.00000	9.80000	60.80000	6.23800	578.00000	105.00000	170.00000	50.00000	0.0	0.0	0.0	0.0	34.2	8.3
1	spring	small	medium	8.35000	8.00000	57.75000	1.28800	370.00000	428.75000	558.75000	1.30000	1.4	7.6	4.8	1.9	6.7	0.0
2	autumn	small	medium	8.10000	11.40000	40.02000	5.33000	346.6									0
3	spring	small	medium	8.07000	4.80000	77.36400	2.30200	98.18									0
4	autumn	small	medium	8.06000	9.00000	55.35000	10.41600	233.7									1

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 200 entries, 0 to 199
```

```
Data columns (total 18 columns):
```

#	Column	Non-Null Count	Dtype
0	Season	200 non-null	str
1	Size	200 non-null	str
2	Speed	200 non-null	str
3	max_pH	200 non-null	str
4	min_O2	200 non-null	str
5	mean_Cl	200 non-null	str
6	mean_NO3	200 non-null	str
7	mean_NH4	200 non-null	str
8	mean_oPO4	200 non-null	str
9	mean_PO4	200 non-null	str
10	mean_Chlor	200 non-null	str
11	a1	200 non-null	float64

Dataframe looks a bit better, but why are numeric columns converted as **object**?
Reading instructions.txt we see that missing values are indicated by XXXXXXXX.

Algae_Blooms — load (4th attempt)

V

```
names = ('Season', 'Size', 'Speed', 'max_pH', 'min_O2', 'mean_Cl', 'mean_NO3', 'mean_NH4', 'mean_oP04',
         'mean_PO4', 'mean_Chlor', 'a1', 'a2', 'a3', 'a4', 'a5', 'a6', 'a7')
df = pd.read_table('src/Analysis.txt', sep=r'\s+', names=names, na_values='XXXXXXX')
print(df.shape)
df.head()
```

(200, 18)

	Season	Size	Speed	max_pH	min_O2	mean_Cl	mean_NO3	mean_NH4	mean_oP04	mean_PO4	mean_Chlor	a1	a2	a3	a4	a5	a6
0	winter	small	medium	8.00	9.8	60.800	6.238	578.00000	105.000	170.00000	50.0	0.0	0.0	0.0	0.0	34.2	8.3
1	spring	small	medium	8.35	8.0	57.750	1.288	370.00000	428.750	558.75000	1.3	1.4	7.6	4.8	1.9	6.7	0.0
2	autumn	small	medium	8.10	11.4	40.020	5.330	346.6									
3	spring	small	medium	8.07	4.8	77.364	2.302	98.18									
4	autumn	small	medium	8.06	9.0	55.350	10.416	233.7									

<class 'pandas.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Season	200 non-null	str
1	Size	200 non-null	str
2	Speed	200 non-null	str
3	max_pH	199 non-null	float64
4	min_O2	198 non-null	float64
5	mean_Cl	190 non-null	float64
6	mean_NO3	198 non-null	float64
7	mean_NH4	198 non-null	float64
8	mean_oP04	198 non-null	float64
9	mean_PO4	198 non-null	float64
10	mean_Chlor	188 non-null	float64
11	a1	200 non-null	float64
12	a2	200 non-null	float64

Now some variables have missing values

Also we should convert Season, Size and Speed to category and ensure the levels are ordered.

Algae_Blooms — Fix Data Types

II

The three categorical variables have levels with a natural order $\Rightarrow$ convert to category and specify order:

```
df.Season = pd.Categorical(df.Season, categories=['spring', 'summer', 'autumn', 'winter'], ordered=True)
print(df.Season.unique())
```

```
['winter', 'spring', 'autumn', 'summer']
Categories (4, str): ['spring' < 'summer' < 'autumn' < 'winter']
```

```
df.Size = pd.Categorical(df.Size, categories=['small', 'medium', 'large'], ordered=True)
print(df.Size.unique())
```

```
['small', 'medium', 'large']
Categories (3, str): ['small' < 'medium' < 'large']
```

```
df.Speed = pd.Categorical(df.Speed, categories=['low', 'medium', 'high'], ordered=True)
print(df.Speed.unique())
```

```
['medium', 'high', 'low']
Categories (3, str): ['low' < 'medium' < 'high']
```

Algae_Blooms — Identification of Missing Values (NA)

1 Which columns have missing values?

```
df.isna().sum()
```

```
Season      0
Size        0
Speed       0
max_pH      1
min_O2      2
mean_Cl     10
mean_NO3    2
mean_NH4    2
mean_oPO4   2
mean_PO4    2
mean_Chlor  12
a1          0
a2          0
a3          0
a4          0
a5          0
a6          0
a7          0
dtype: int64
```

- Two columns (features) account for 22 NAs, but cannot just drop them as will lose a lot of information.
- Two rows (observations) account for 12 NAs $\Rightarrow$ remove.
- Removing other rows with a NA will result in a loss of 14 rows (7% of the data), instead will impute later.

2 Which rows have missing values?
How many NAs per row?

```
df.isna().sum(axis=1).value_counts()
```

```
0    184
1      7
2      7
6      2
Name: count, dtype: int64
```

4 Rows / Cols to drop?

```
df.loc[df.isna().sum(axis=1)==6]
```

	Season	Size	Speed	max_pH	min_O2	mean_Cl	mean_NO3	mean_NH4	mean_oPO4	mean_PO4	mean_Chlor	a1	a2	a3
61	summer	small	medium	6.4	NaN	NaN	NaN	NaN	NaN	14.0	NaN	19.4	0.0	0.0
198	winter	large	medium	8.0	7.6	NaN	NaN	NaN	NaN	NaN	NaN	0.0	12.5	3.0

```
df = df.loc[df.isna().sum(axis=1)<6].copy()
print(df.shape)
```

```
(198, 18)
```

Handling missing values using pandas

- Step 1: Replace missing values with marked values. Examples:
`df[col].replace('XXXXXX', np.nan)` or `df[col].replace("", None)`
- Step 2: Count the missing values by row and by column.
- Step 3: If a row or a column has **too many(?)** missing values, either drop it or add an extra boolean-valued feature `df[hasCol]`
- Step 4: If future processing can handle data with missing values, skip to step 7.
- Step 5: Otherwise, replace the missing values with **representative(?)** values...
- Step 5a: If the data is categorical, use a **safe** value such as its mode (the most commonly occurring value)
- Step 5b: If the column is numeric, unsorted, symmetrically distributed, without outliers, try
`df[col].fillna(df[col].mean(), inplace=True)`
- Step 5c: If the column is numeric, unsorted, but is unsymmetrically distributed and/or has outliers, try
`df[col].fillna(df[col].median(), inplace=True)`
- Step 6: If the column is numeric and sorted by row index, `df[col].interpolate()` can fill in values, so 1, 3, , 7, 8 becomes 1, 3, 5, 7, 8.
- Step 7: Proceed to EDA Pass 2

After Loading and Initial Clean — Where are we?

Tips

- ✓ Loaded data, corrected dtypes (categorical with order levels)
- ✓ Sanitised column names — not needed, but note column name size shadows pandas dataframe function size $\Rightarrow$ so use `df["size"]` instead of `df.size`.
- ✓ No missing values

Titanic

- ✓ Loaded data — no conversion of dtypes needed (but if you don't plots/crosstab order won't agree)
- ✓ Sanitised column names — not needed,
 - Missing values in Age (177/891=20%), Cabin (687/891=77%), and Embarked (2/891=0.2%).
 - A feature with 77% missing values should be considered for deletion, but what if the presence of a missing value actually tells us something? $\Rightarrow$ convert to a boolean feature.

Algae Blooms

- ✓ Loaded data, corrected dtypes (categorical with ordered levels)
- ✓ Sanitised column names.
 - Missing values
 - Removed two rows with 6 NA each, accounted for 12/33=36% of the missing values.
 - Remaining, 21 NAs are concentrated in mean_CL (8) and mean_Chlor (10). EDA will suggest options.

After Loading and Initial Clean — Where are we?

Next we might

- Save result of initial clean:
 - To either a CSV (if we don't mind losing dtype metadata)

```
df.to_csv('data/Analysis.csv', index=False)
```

- To (say) pickle format (to keep dtype metadata)

```
df.to_pickle('data/Analysis.pkl')
```

Later can read dataframe back in using

```
df = pd.read_pickle('data/Analysis.pkl')  
print(df.shape)  
df.head(1)
```

- If the dataset is large (>100K rows), save a (reproducible) sample of the dataset for later EDA to speed up calculations (especially visualisations).

```
df.sample(frac=.25, random_state=42).to_pickle('data/Analysis_sample.pkl')
```

End of Pass 1...